

	NOTA TECNICA TECHNICAL REPORT		Núm. No.	NT-T-ADP-08005
	Departamento Department	Methods	Avión Aircraft	Technologies
Título/Title Program PROCESASEC, Version 2 User's Manual				
<u>Palabras clave/Key words</u> MANUALS / STRESS ANALYSIS / COMPUTER PROGRAMS			Clasificación acceso Access class P1	Registro de Revisiones Revision Record Pag. 2
<u>Resumen/Summary</u> The purpose of this Technical Report is to define the way to use the PROCESASEC program. This program processes a flight-by-flight, cycle-by-cycle random stress sequence. The input sequence can be provided in two different ways: - Sequence of discrete flights. - Continuous stress recording. The processed sequence will be generated with the following particularities: - The stress sequence is filtered, removing all intermediate stress states. - The sequence can be modified by applying a factor on stresses and adding a constant stress level. - Small cycles can be removed from the final sequence by defining an appropriate omission level. - A Rainflow counting method can be applied on the sequence. - A block spectrum is also generated from the final stress sequence. The final stress sequence is written in binary format and it can be used, as it is, as input for FATIGA and GRIETA programs. PROPIEDAD DE EADS CASA, Sociedad Unipersonal. Este documento no puede ser utilizado ni reproducido total o parcialmente sin la previa autorización escrita de la Dirección de Ingeniería y Tecnología de EADS-CASA. EADS CASA, Sociedad Unipersonal PROPERTY. This document shall neither be used nor completely or partially reproduced without previous written authorization by EADS-CASA Engineering and Technology Directorate.				
Distribución/Distribution	Realización/Execution		Fecha 1ª edición 1 st issue Date July -2008	
Archivo de Cálculo EADS CASA	Realizado Prepared		Firma Signature	
Subdirector de Cálculo (*)	Darío Fernández Villalba			
Jefes de Depto. de Cálculo (*)				
Jefes de Sección de Cálculo				
	Comprobado Checked	Lucas Claramonte Clausell		
(*).- Cover sheet only	Aprobado Approved	Darío Fernández Villalba		

REGISTRO DE REVISIONES/REVISIONS RECORD

[illegible]

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1.- Technical sheet of the program

Technical Note **NT-T-ADP-08005**PROGRAM

PROCESASEC

DATE

2008

OBJECTIVE

This program processes a flight-by-flight, cycle-by-cycle random stress sequence to be used for further applications.

LANGUAGE

Fortran 77

COMPUTER

Sun Solaris

NUMBER OF LINES (APROX.)

2 000

EXECUTION MODE

Interactive

AUTHOR

Dario Fernández Villalba

VERSION

2.0

ADDITIONAL DOCUMENTS

Technical Note: NT-T-ADP-08006.
Theoretical Manual of the PROCESASEC Program, Version 2.

Technical Note: NT-T-ADP-08007.
Programmer's Manual of the PROCESASEC Program, Version 2.

2.- Introduction

The purpose of this Technical Report is to define the way to use the PROCESASEC program.

This program processes a flight-by-flight, cycle-by-cycle random stress sequence. The input sequence can be provided in two different ways:

- Sequence of discrete flights.
- Continuous stress recording.

The processed sequence will be generated with the following particularities:

- The stress sequence is filtered, removing all intermediate stress states.
- The sequence can be modified by applying a factor on stresses and adding a constant stress level.
- Small cycles can be removed from the final sequence by defining an appropriate omission level.
- A Rainflow counting method can be applied on the sequence.
- A block spectrum is also generated from the final stress sequence.

The final stress sequence is written in binary format and it can be used, as it is, as input for FATIGA and GRIETA programs.

Differences Between versions:

- Version 1.0:

Initial version developed in Fortran 77.

In parallel, a Windows application has been developed as Graphic User Interface GUI for the program. This application uses the same input file and provides the same stress sequence and summary. The process is performed by calling to the Dynamic Link Library DLL, compiled from the same source code as the official Fortran version.

- Version 1.1:

Some irrelevant changes are included in the Fortran version and the Graphic User Interface is improved.

- Version 1.2:

Small changes in the procedure to filter and depurate the sequence are included to avoid the loss of some relevant stress level at the beginning or at the end of the sequence.

The Graphic User Interface is improved and adapted to the new platform Microsoft .NET Framework.

- Version 1.3:

It is solved the error produced for input sequences, given in terms of continuous stress recording, with a number of stress states exceeding the program capability

- Version 1.4:

It is solved the error produced when attempting to apply the Rainflow on a sequence having a unique cycle. In such case, the Rainflow is not applied.

- Version 2.0:

Now it is possible to apply a factor, positive or negative, on all stresses in the sequence and to add a constant stress level, as follows:

$$\sigma = \sigma_{\text{add}} + \text{Factor} \cdot \sigma_o$$

3.- Execution of the program

Prior to the program execution, it is necessary to create the input data file, as described in section 4. The program can be then executed by typing its **full path name**.

If the software symbol table has been previously assigned (consult with the System Manager in order to know how to assign this symbol table), it can be executed by typing the command:

```
$procesasec
```

The program asks for the language to use, the parameters defining the process and the name of the input and output files, as shown in the following example:

```
Idioma / Language:
=====

  1 ... ESPAÑOL
  2 ... ENGLISH
Idioma / Language: 2
Type of sequence:
=====

  1 ... Sequence of discrete flights
  2 ... Continuous stress recording
Select the type of sequence: 1
Type of Rainflow:
=====

  0 ... Without Rainflow
  1 ... Local Rainflow
  2 ... Local Rainflow + GAG
  3 ... Full Rainflow
  4 ... Full Rainflow rearranged
Select the type of rainflow to apply: 4
Omission level: 10.0
Additional stress: 10.0
Factor on stresses: 1.15
```

Name of the input file: Example.dat

Name of the output file for the sequence: Example.cbc

Name of the output file for the summary: Example.sum

The CPU time for calculation depends on the size of the input sequence, being a few seconds as much.

It is possible to execute the program with a single command, including the input and output filenames, process data and the language as parameters. The above example can be executed as follows:

```
$procesasec Example.dat Example.cbc Example.sum 1 4 10.0 10.0 1.15 2
```

The required parameters to be used in the above command, must be included in the following order:

- Input sequence filename
- Output sequence filename
- Summary filename
- Type of input sequence (1 or 2)
- Type of Rainflow counting method (0 to 4)
- Omission level
- Additional stress
- Factor on stresses, positive or negative, but not zero
- Language (1 or 2)

4.- Input sequence file

This file contains the input sequence to be processed by the program. The arrangement of this file depends on the type of sequence.

4.1 Sequence of discrete flights

In this case, the input stress sequence is defined as a succession of flights, each one of them containing several stress states. For this kind of sequence, the organisation of the input file is the following:

```

Nflt
  Nst(1)
    Stress(1,1)
    Stress(1,2)
    .....
    .....
    Stress(1,Nst(1))
  Nst(2)
    Stress(2,1)
    Stress(2,2)
    .....
    .....
    Stress(2,Nst(2))
  .....
  .....
  .....
  Nst(Nflt)
    Stress(Nflt,1)
    Stress(Nflt,2)
    .....
    .....
    Stress(Nflt,Nst(Nflt))

```

} Flight 1
 } Flight 2
 } Flight Nflt

The meaning of each one of the above parameters is the following:

- Nflt: Is the number of flights in the sequence.
- Nst(i): Is the number of stress states contained in the flight “i”.
- Stress(i,j): Is the stress for the state “j” in the flight “i”.

Stresses contained in the same flight can be written grouped in the same line.

4.2 Continuous stress recording

In this case, the input stress sequence is defined as a succession of stress states, without any indication of beginning or end of flight. For this kind of sequence, the organisation of the input file is the following:

```
Code(1)   Stress(1)
Code(1)   Stress(2)
.....
.....
Code(n)   Stress(n)
```

The meaning of each one of the above parameters is the following:

- Code(i): Is a number that can be used to identify the origin of the stress state. Nevertheless the program ignores this value
- Stress(i): Is the stress in the state “i” of the sequence.

5.- Output results

The PROCESASEC program creates two files, one of them is not printable, with the flight-by-flight, cycle-by-cycle random stress sequence, and the other is a printable file, containing the summary of the processed sequence:

- Output stress sequence file.
- Output summary file.

5.1 Output stress sequence file

This file contain the flight by flight, cycle by cycle stress sequence and it is written in binary form, compatible with other programs, like GRIETA (input file), FATIGA (input file) or SECUENCIA (output file).

The first record has the number of flights in the sequence, and the subsequent records have a complete flight each one.

This file can be read by means of the following sub program (written in Fortran):

```
*****
* DECLARATIONS *
*****
      INTEGER*4 NFL,NCYC(xx),IFL,ICYC
      REAL*4  SMIN(xx,yy),SMAX(xx,yy)

*****
* FILE OPENING *
*****
      OPEN (1,FILE='filename',STATUS='OLD',FORM='UNFORMATTED')

*****
* TOTAL NUMBER OF FLIGHTS *
*****
      READ (1) NFL

*****
* LOOP OF FLIGHTS *
*****
      DO IFL=1,NFL

*****
* READING OF A FLIGHT *
*****
      READ (1) IFL,NCYC(IFL),(SMIN(IFL,ICYC),SMAX(IFL,ICYC),ICYC=1,NCYC(I))

*****
* END OF LOOP OF FLIGHTS *
*****
      END DO

*****
* FILE CLOSING *
*****
      CLOSE(1)
```

The meaning of each one of the above variables is the following:

NFL..... Is the total number of flights in the sequence.

NCYC(IFL) Is the number of cycles in the flight number “IFL”.

SMIN(IFL,ICYC)..... Is the minimum of the cycle number “ICYC” in the flight number “IFL”.

SMAX(IFL,ICYC) Is the maximum of the cycle number “ICYC” in the flight number “IFL”.

IFL and ICYC..... Are loop variables.

When the input stress sequence is defined as a sequence of discrete flights, the number of flights in the output sequence “NFL” is the same as for the input sequence.

If the input stress sequence is defined as a continuous recording, the number of flights in the output sequence will be $NFL = 1$.

5.2 Output summary file

This file includes the following information:

- **Block spectrum:**

The block spectrum contains all cycles of the output stress sequence, grouped in blocks of cycles. It can be used as input spectrum for the program GRIETA. It has one line for each block, in the following way:

Maximum stress	Minimum stress	Number of cycles of the block
Maximum stress	Minimum stress	Number of cycles of the block
.....		
.....		
Maximum stress	Minimum stress	Number of cycles of the block

- **Summary of the random stress sequence:**

This section of the file contains the following data:

- Table of the peaks distribution, in the form of an interval of peak stresses versus the frequency.
- Table of the amplitude distribution, given in the same way.
- Table of the valley-peak ratio distribution, also given in the same way.
- A summary showing:
 - Maximum peak
 - Minimum valley
 - Maximum range
 - Number of total cycles
 - Number of total flights
 - Number of cycles per flight

Examples of the output summary file are provided in chapter 7.

When the program detects an error in the input data, it writes the corresponding message in the console and stops the execution. The error messages that might appear are listed on next page:

Error messages. English output:

ERROR: The input file cannot be opened
ERROR: The output file for the sequence cannot be opened
ERROR: The output file for the summary cannot be opened
ERROR: Invalid type of sequence
ERROR: Invalid type of Rainflow
ERROR: Invalid omission level
ERROR: Invalid additional stress
ERROR: Invalid factor on stresses
ERROR: Bad input data in input sequence file
ERROR: End of file while reading the input sequence file

Error messages. Spanish output:

ERROR: No se puede abrir el fichero de entrada
ERROR: No se puede abrir el fichero de salida para la secuencia
ERROR: No se puede abrir el fichero de salida para el sumario
ERROR: Tipo de secuencia incorrecto
ERROR: Tipo de "Rainflow" incorrecto
ERROR: Nivel de omisión incorrecto
ERROR: Esfuerzo adicional incorrecto
ERROR: Factor sobre los esfuerzos incorrecto
ERROR: Datos erróneos en el fichero de la secuencia de entrada
ERROR: Fin de fichero mientras se leía la secuencia de entrada

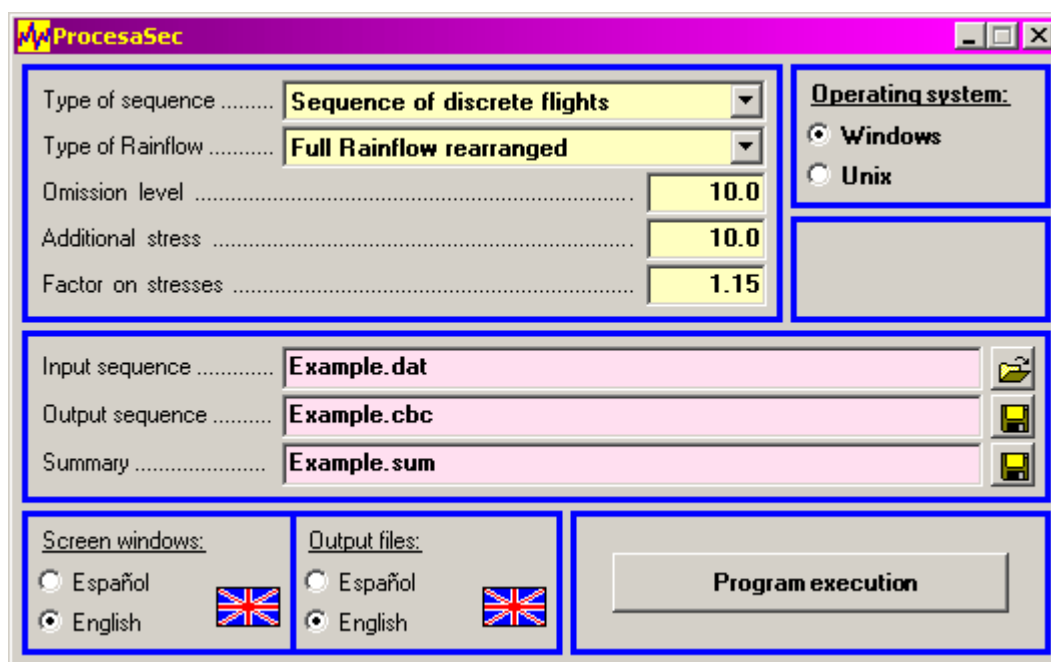
6.- Windows application

An application has been developed in Visual Basic for Windows environment. The original routines of the program for calculation and output results have been included in a Dynamic Link Library, which is used by this application. Therefore the same results will be obtained, with only differences due to rounding.

Since this is a Windows application, it must be installed in the user's PC before it can be executed. For installation, please consult with the system manager.

The application can be executed by “double click” in the file **ProcesaSec.exe**, located in the PC's server, or by means a **link** in the “desktop” or in the “start menu”.

The input data must be typed in the application form. This form, after typing the data corresponding to the example 1 (see section 7.1), has the following appearance:



The screenshot shows the ProcesaSec application window. It contains several input fields and buttons. The 'Type of sequence' is set to 'Sequence of discrete flights'. The 'Type of Rainflow' is set to 'Full Rainflow rearranged'. The 'Omission level' is 10.0, 'Additional stress' is 10.0, and 'Factor on stresses' is 1.15. The 'Operating system' is set to 'Windows'. The 'Input sequence' is 'Example.dat', 'Output sequence' is 'Example.cbc', and 'Summary' is 'Example.sum'. The 'Screen windows' and 'Output files' sections both have 'Español' and 'English' options, with 'English' selected. A 'Program execution' button is at the bottom right.

Figure 1: Input data form.

In the above form, the following data have been specified:

- The input sequence is provided as a sequence of discrete flights.
- A “full Rainflow rearranged” will be applied on the stress sequence, with a constant omission level of 10 N/mm^2 (assuming that the input sequence is also expressed in N/mm^2).
- Before the above process, a factor of 1.15 will be applied on the original sequence and a constant stress level of 10.0 will be added.
- The output stress sequence will be generated as a binary file for Microsoft Windows operating system. Therefore it cannot be used as input for applications installed in Unix.

Note that it is also possible to generate the sequence as a binary file for a computer with Unix operating system by selecting the Unix option in the above form. In such a case, the sequence could be used as input for applications installed in Unix machines, but not for Windows.

The program can be then executed by clicking the button “Program execution”. The application starts to solve the problems showing the following form at the end:

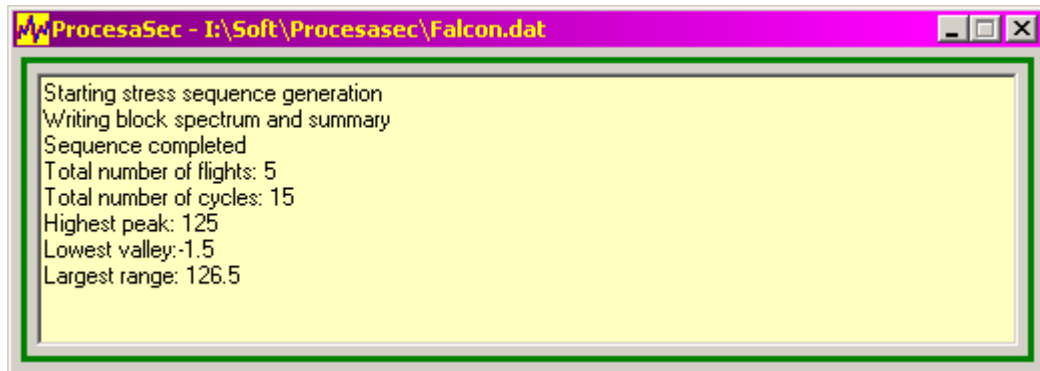


Figure 2: Tracking and results form.

The output results are written in the output files ***Example.cbc*** (binary file with the random stress sequence) and ***Example.sum*** (printable file with the sample, the block spectrum and the summary).

7.- Examples

Two examples of how to use the program are given below. Each one of them corresponds to a different input sequence format.

7.1 Example 1

- Input stress sequence:

It is a sequence of five discrete flights. All them are equal, with the following sequence of stresses each one, expressed in N/mm^2 :

60.0 40.0 20.0 50.0 80.0 60.0 40.0 70.0 100.0 45.0 -10.0 -5.0 0.0

- Output stress sequence:

The above stresses will be multiplied by a factor of 1.15 and, after that, a constant stress level of 10 N/mm^2 will be added.

It will be generated applying a “Full Rainflow Rearranged” counting method with an omission level of 10 N/mm^2 in range.

The input sequence as well as the output summary file are provided on next pages:

Input stress sequence file:

5

13

60.0	40.0	20.0	50.0	80.0
60.0	40.0	70.0	100.0	45.0
-10.0	-5.0	0.0		

13

60.0	40.0	20.0	50.0	80.0
60.0	40.0	70.0	100.0	45.0
-10.0	-5.0	0.0		

13

60.0	40.0	20.0	50.0	80.0
60.0	40.0	70.0	100.0	45.0
-10.0	-5.0	0.0		

13

60.0	40.0	20.0	50.0	80.0
60.0	40.0	70.0	100.0	45.0
-10.0	-5.0	0.0		

13

60.0	40.0	20.0	50.0	80.0
60.0	40.0	70.0	100.0	45.0
-10.0	-5.0	0.0		

Output summary file. Block spectrum:

***** B L O C K S P E C T R U M *****

Smax	Smin	Cycles
----	----	-----
77.50	34.88	5
102.50	56.38	5
127.50	-6.38	5

Output summary file. Sequence summary:

***** O V E R A L L S E Q U E N C E S U M M A R Y *****

Peak distribution			Range distribution			Valley/Peak distribution		
=====			=====			=====		
0.00	5.00	0	0.00	5.00	0	-1.5	-1.4	0
5.00	10.00	0	5.00	10.00	0	-1.4	-1.3	0
10.00	15.00	0	10.00	15.00	0	-1.3	-1.2	0
15.00	20.00	0	15.00	20.00	0	-1.2	-1.1	0
20.00	25.00	0	20.00	25.00	0	-1.1	-1.0	0
25.00	30.00	0	25.00	30.00	0	-1.0	-0.9	0
30.00	35.00	0	30.00	35.00	0	-0.9	-0.8	0
35.00	40.00	0	35.00	40.00	0	-0.8	-0.7	0
40.00	45.00	0	40.00	45.00	0	-0.7	-0.6	0
45.00	50.00	0	45.00	50.00	10	-0.6	-0.5	0
50.00	55.00	0	50.00	55.00	0	-0.5	-0.4	0
55.00	60.00	0	55.00	60.00	0	-0.4	-0.3	0
60.00	65.00	0	60.00	65.00	0	-0.3	-0.2	0
65.00	70.00	0	65.00	70.00	0	-0.2	-0.1	0
70.00	75.00	0	70.00	75.00	0	-0.1	0.0	5
75.00	80.00	5	75.00	80.00	0	0.0	0.1	0
80.00	85.00	0	80.00	85.00	0	0.1	0.2	0
85.00	90.00	0	85.00	90.00	0	0.2	0.3	0
90.00	95.00	0	90.00	95.00	0	0.3	0.4	0
95.00	100.00	0	95.00	100.00	0	0.4	0.5	5
100.00	105.00	5	100.00	105.00	0	0.5	0.6	5
105.00	110.00	0	105.00	110.00	0	0.6	0.7	0
110.00	115.00	0	110.00	115.00	0	0.7	0.8	0
115.00	120.00	0	115.00	120.00	0	0.8	0.9	0
120.00	125.00	0	120.00	125.00	0	0.9	1.0	0
125.00	130.00	5	125.00	130.00	5	1.0	1.1	0
130.00	135.00	0	130.00	135.00	0	1.1	1.2	0
135.00	140.00	0	135.00	140.00	0	1.2	1.3	0
140.00	145.00	0	140.00	145.00	0	1.3	1.4	0
145.00	150.00	0	145.00	150.00	0	1.4	1.5	0
150.00	155.00	0	150.00	155.00	0			
155.00	160.00	0	155.00	160.00	0	Maximum peak	=	125.00
160.00	165.00	0	160.00	165.00	0	Minimum valley	=	-1.50
165.00	170.00	0	165.00	170.00	0	Maximum range	=	126.50
170.00	175.00	0	170.00	175.00	0			
175.00	180.00	0	175.00	180.00	0	Cycles:		15
180.00	185.00	0	180.00	185.00	0	Flights:		5
185.00	190.00	0	185.00	190.00	0	Cycles/Flight:		3.000

7.2 Example 2

- Input stress sequence:

It is a continuous stress recording with 263086 stress states.

- Output stress sequence:

All stresses in the input sequence will be multiplied by a factor of 1.15. No constant stress level will be added.

It will be generated applying a “Full Rainflow Rearranged” counting method without any omission level.

The input sequence as well as the output summary file are provided on next pages:

Input stress sequence file:

1048	-0.092
1048	10.690
1048	29.140
1048	28.010
1048	10.690
1048	21.472
1048	10.690
1008	4.555
1008	10.690
1019	14.079
1019	10.690
1019	29.140
1019	28.010
1019	10.690
1019	7.301
1019	10.690
1003	4.555
1003	10.690
1035	17.159
1035	10.690
1035	40.830
1035	45.430
1035	10.690
1035	4.221
1035	10.690
1003	4.555
1003	10.690
1036	4.221
1036	10.690
1036	40.830
1036	45.430
1036	10.690
1036	17.159
1036	10.690
1003	4.555
1003	10.690
1009	29.140
1009	28.010
1009	10.690
1003	4.555
1003	10.690
1019	7.301
1019	10.690
1019	29.140
1019	28.010
1019	10.690
1019	14.079
1019	10.690
1003	4.555
....	
....	
....	
....	

Output Summary file. Block spectrum:

***** B L O C K S P E C T R U M *****

Smax	Smin	Cycles
----	----	-----
12.50	-0.62	68
12.50	4.38	246
12.50	5.62	1437
12.50	6.88	314
12.50	8.12	7838
17.50	-0.88	39
17.50	4.38	41
17.50	7.88	5
17.50	9.62	42
17.50	11.38	1404
17.50	13.12	6904
22.50	-1.12	1
22.50	3.38	6
22.50	7.88	1
22.50	10.12	497
27.50	6.88	52
27.50	9.62	1
27.50	12.38	1
27.50	15.12	220
27.50	17.88	2729
27.50	20.62	10762
32.50	-14.63	5
32.50	-11.38	1
32.50	-1.62	14
32.50	4.88	4470
32.50	8.12	472
32.50	11.38	55
32.50	14.62	16
32.50	21.12	113
32.50	27.62	9630
32.50	30.88	254
37.50	-1.88	3
37.50	5.62	93
37.50	9.38	1
37.50	13.12	10
37.50	20.62	12
37.50	24.38	1927
42.50	-14.88	5
42.50	-2.12	11
42.50	6.38	14
47.50	-16.63	1
47.50	-11.88	12
47.50	-2.38	9
47.50	2.38	3
47.50	7.13	225
47.50	16.62	3
47.50	21.38	12
47.50	26.12	699
47.50	40.38	6
47.50	45.12	52

Output Summary file. Block spectrum: (cont.)

***** B L O C K S P E C T R U M *****

Smax	Smin	Cycles
----	----	-----
52.50	-18.38	1
52.50	-13.12	25
52.50	-2.62	155
52.50	2.62	612
52.50	7.88	2557
52.50	13.12	30
52.50	18.38	1
52.50	44.62	8
52.50	49.88	12
57.50	-14.38	83
57.50	-2.88	11
57.50	2.88	31
57.50	8.62	24
57.50	14.38	731
57.50	20.12	10418
57.50	54.62	72
62.50	-21.88	2
62.50	-15.62	173
62.50	-9.38	6
62.50	-3.12	239
62.50	3.12	1817
62.50	9.38	7
62.50	15.62	6
67.50	-16.88	19
67.50	-10.12	1
67.50	10.12	3
67.50	16.88	75
72.50	10.88	2
72.50	18.12	17
77.50	-11.62	107
77.50	3.88	154
77.50	11.62	460
77.50	19.38	6323
77.50	73.62	182
82.50	-12.38	4
82.50	-4.12	7
82.50	4.12	45
82.50	12.38	27
82.50	20.62	268
87.50	-21.88	1
87.50	-13.13	291
87.50	-4.38	10
87.50	4.38	239
87.50	13.13	2147
87.50	21.88	26
87.50	83.12	100
92.50	-23.12	1
92.50	-13.88	204
92.50	-4.62	15
92.50	4.62	273

Output Summary file. Block spectrum: (cont.)

***** B L O C K S P E C T R U M *****

Smax	Smin	Cycles
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92.50	13.88	345
92.50	23.12	13
92.50	87.88	35
97.50	-14.63	162
97.50	-4.88	2
97.50	4.88	51
97.50	14.63	115
97.50	24.38	1
102.50	-25.62	5
102.50	-15.38	14
102.50	5.12	4
102.50	15.38	3
107.50	-26.88	1
107.50	-16.12	2

Output summary file. Sequence summary:

***** O V E R A L L S E Q U E N C E S U M M A R Y *****

Peak distribution			Range distribution			Valley/Peak distribution		
=====			=====			=====		
0.00	5.00	0	0.00	5.00	15433	-1.5	-1.4	0
5.00	10.00	0	5.00	10.00	26594	-1.4	-1.3	0
10.00	15.00	9903	10.00	15.00	2871	-1.3	-1.2	0
15.00	20.00	8435	15.00	20.00	827	-1.2	-1.1	0
20.00	25.00	505	20.00	25.00	66	-1.1	-1.0	0
25.00	30.00	13765	25.00	30.00	4955	-1.0	-0.9	0
30.00	35.00	15030	30.00	35.00	114	-0.9	-0.8	0
35.00	40.00	2046	35.00	40.00	10408	-0.8	-0.7	0
40.00	45.00	30	40.00	45.00	1138	-0.7	-0.6	0
45.00	50.00	1022	45.00	50.00	3160	-0.6	-0.5	0
50.00	55.00	3401	50.00	55.00	226	-0.5	-0.4	5
55.00	60.00	11370	55.00	60.00	2120	-0.4	-0.3	10
60.00	65.00	2250	60.00	65.00	6885	-0.3	-0.2	320
65.00	70.00	98	65.00	70.00	268	-0.2	-0.1	791
70.00	75.00	19	70.00	75.00	2554	-0.1	0.0	584
75.00	80.00	7226	75.00	80.00	366	0.0	0.1	3229
80.00	85.00	351	80.00	85.00	583	0.1	0.2	10498
85.00	90.00	2814	85.00	90.00	96	0.2	0.3	8056
90.00	95.00	886	90.00	95.00	164	0.3	0.4	10735
95.00	100.00	331	95.00	100.00	4	0.4	0.5	1968
100.00	105.00	26	100.00	105.00	459	0.5	0.6	1287
105.00	110.00	3	105.00	110.00	141	0.6	0.7	14011
110.00	115.00	0	110.00	115.00	53	0.7	0.8	17666
115.00	120.00	0	115.00	120.00	15	0.8	0.9	9644
120.00	125.00	0	120.00	125.00	8	0.9	1.0	707
125.00	130.00	0	125.00	130.00	3	1.0	1.1	0
130.00	135.00	0	130.00	135.00	0	1.1	1.2	0
135.00	140.00	0	135.00	140.00	0	1.2	1.3	0
140.00	145.00	0	140.00	145.00	0	1.3	1.4	0
145.00	150.00	0	145.00	150.00	0	1.4	1.5	0
150.00	155.00	0	150.00	155.00	0			
155.00	160.00	0	155.00	160.00	0	Maximum peak	=	106.86
160.00	165.00	0	160.00	165.00	0	Minimum valley	=	-23.03
165.00	170.00	0	165.00	170.00	0	Maximum range	=	129.89
170.00	175.00	0	170.00	175.00	0			
175.00	180.00	0	175.00	180.00	0	Cycles:		79511
180.00	185.00	0	180.00	185.00	0	Flights:		1
185.00	190.00	0	185.00	190.00	0	Cycles/Flight:		79511.000